

Design and Development of a USB-Powered Secure File Transfer Authentication Module Using Arduino

Abstract

This project presents the design and development of a USB-powered secure file transfer authentication module using an Arduino microcontroller. The system provides an additional layer of security by authenticating authorized users before permitting file transfer operations between connected devices. Powered entirely through a standard 5 V USB interface, the module is portable, energy-efficient, and easy to deploy. User authentication can be implemented using RFID or password-based verification, enabling the Arduino to grant or deny access based on stored credentials. Visual and audible indicators inform users of successful or failed authentication attempts, while repeated unauthorized access attempts may trigger a security alert. The proposed design offers a low-cost and reliable solution for protecting sensitive data during file transfers in educational, office, and personal environments. Future enhancements may include biometric authentication, encrypted communication, and wireless connectivity for improved security and remote management.